A Lean-Certified Endpoint-Shape Counterexample in the Branch-C Farey Mediator Program

Why family-level coverage cannot be promoted to universal coverage without a boundary audit

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Abstract

A proof program can establish a sequence of valid results for several parameterized families and still fail when those families are promoted to a universal classification. This paper isolates that failure inside the finite Branch-C Farey mediator program. The proposed contract `arbitraryEndpointShape BridgeContract` asserted that every bounded, reduced, strictly ordered endpoint pair with determinant greater than one belonged to the then-current union of endpoint-shape families. The pair

$$N = 3, \qquad a = (0, 1), \qquad b = (2, 3) \tag{1}$$

is a counterexample. It satisfies the contract's complete premise, including determinant 2, but it is outside the enumerated shape union. The resulting negation of the bridge is machine-checked in Lean 4.30.0. This revision adds a pinned, hash-governed replay bundle and a second proof layer that avoids `native_decide` in the four essential results. The dependency report for those new results contains only Lean's standard `propext` and `Quot.sound` foundations. A further regression theorem proves that the same pair has the listed mediator $(1, 2)$. The result therefore separates two propositions that must not be conflated: the endpoint-shape coverage bridge is false, while the listed-mediator target may remain true through a different construction. The paper preserves the original counterexample and makes its problem, proof, logical consequence, later bypass, and reproducibility boundary explicit.

Scope and evidence boundary

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1. The problem before the terminology

Suppose a proof program has verified several families of cases. Each family may be correct. The union of those families may still omit a boundary case. If the program then asserts that the union covers every admissible input, the new universal statement is a separate theorem, not a consequence of the family proofs alone.

This distinction is easy to lose in a long formal-development chain. Local lemmas accumulate, a recurring endpoint pattern becomes familiar, and the gap between "every member of these families works" and "every admissible endpoint belongs to one of these families" can become implicit. The right response is to name that promotion step as its own proposition and test it.

The Branch-C program reached exactly this point. It had proved several finite endpoint constructions, including the four-plus-one family. What remained was a proposed bridge from those family results to all canonical endpoint pairs with determinant greater than one. Once the bridge was written as an explicit Lean proposition, a small zero-left boundary case refuted it.

The contribution of this paper is therefore not a new name for a familiar Farey fact. It is an exact audit of a theorem dependency:

1. identify the domain that a universal bridge claims to cover;
2. identify the union of families claimed to cover it;
3. exhibit an admissible pair outside that union;
4. machine-check the premise, the exclusion, and the resulting negation;
5. keep that negation separate from any stronger target that might be proved by another route.

2. Established setting and local formalization

Farey sequences consist of reduced rational numbers with bounded denominators, arranged in increasing order. Their arithmetic and subsequence structure are standard subjects in elementary and combinatorial number theory [1]. The present formalization uses natural-number pairs rather than a quotient type for rational numbers:

$$\mathrm{RawPair} := \mathbb{N} \times \mathbb{N}. \tag{2}$$

For a finite denominator bound N, the local list `rawPairsCanonical N` contains pairs (p, q) satisfying

$$1 \leq q \leq N, \qquad p \leq q, \qquad \gcd(p, q) = 1. \tag{3}$$

The Boolean cross-multiplication relation is

$$\mathrm{fareyLeRaw}(a, b) \ :\Longleftrightarrow\ a_1 b_2 \leq b_1 a_2. \tag{4}$$

The strict relation `fareyStrict a b` requires Equation (4) in the forward direction and rejects it in the reverse direction. For pairs satisfying that strict order, the local determinant is

$$\mathrm{fareyDet}(a, b) := b_1 a_2 - a_1 b_2. \tag{5}$$

Lean represents the subtraction in Equation (5) in $\mathbb{N}$. The strict-order premise makes the intended counterexample positive, so no truncated-subtraction ambiguity affects the calculation used here.

The formal premise `canonicalDetGtOneEndpointPair N a b` joins four conditions:

$$\begin{aligned} \mathrm{EndpointPremise}_N(a, b) \ :\Longleftrightarrow\ & a \in \mathrm{rawPairsCanonical}(N) \\ & \wedge\ b \in \mathrm{rawPairsCanonical}(N) \\ & \wedge\ \mathrm{fareyStrict}(a, b) \\ & \wedge\ 1 < \mathrm{fareyDet}(a, b). \end{aligned} \tag{6}$$

Only after these ordinary objects are fixed do the local Branch-C names become useful. The predicate

`baseOffsetNonOffsetLeftFourPlusOneEndpointShape N a b` is the union of the endpoint families available at the BC-054IR stage. The proposed universal bridge was

$$\mathrm{Bridge} \;:\Longleftrightarrow\; \forall N, a, b, \quad \mathrm{EndpointPremise}_N(a, b) \Rightarrow \mathrm{ShapeUnion}_N(a, b). \tag{7}$$

Equation (7) is the plain-language promotion step written as the Lean definition `arbitraryEndpointShapeBridgeContract`.

3. The counterexample

Take the values in Equation (1). Both endpoints are reduced and bounded at $N = 3$:

$$(0, 1), (2, 3) \in \mathrm{rawPairsCanonical}(3). \tag{8}$$

Cross multiplication gives

$$0 \cdot 3 < 2 \cdot 1, \tag{9}$$

so $(0, 1)$ precedes $(2, 3)$ in the local strict Farey order. The determinant is

$$\mathrm{fareyDet}\big((0, 1), (2, 3)\big) = 2 \cdot 1 - 0 \cdot 3 = 2 > 1. \tag{10}$$

Equations (8)-(10) establish the complete premise of Equation (7). The remaining question is whether the pair belongs to the then-current shape union.

It does not. Each branch of the union conflicts with the zero left numerator. Some families force that numerator to equal 1. Other branches obtain the left numerator from a parameter p whose admissibility conditions force $p > 0$. Substituting $a = (0, 1)$ into each branch therefore produces an arithmetic contradiction. Lean checks the complete case split and proves

$$\neg\, \mathrm{ShapeUnion}_3\big((0, 1), (2, 3)\big). \tag{11}$$

Combining the premise with Equation (11) gives the central result:

$$\neg\, \mathrm{Bridge}\,. \tag{12}$$

Theorem 1. Endpoint-shape bridge counterexample

The then-current `arbitraryEndpointShapeBridgeContract` is false. The witness is $N = 3$, $a = (0, 1)$, $b = (2, 3)$.

Proof

Equation (10), together with canonical membership and strict order, proves `canonicalDetGtOneEndpointPair 3 (0,1) (2,3)`. Equation (11) proves the negation of `baseOffsetNonOffsetLeftFourPlusOneEndpointShape 3 (0,1) (2,3)`. Applying the proposed bridge to that admissible pair contradicts Equation (11). The machine-checked theorem is endpoint_shape_bridge_refuted_without_native_decide. QED.

4. What the counterexample does and does not decide

A coverage bridge and a mediator-existence target are different propositions.

The bridge says that every admissible endpoint pair belongs to one of the enumerated shape families. A listed-mediator target says that some canonical point lies strictly between the endpoints. The first statement can fail even when the second statement is true.

For the counterexample itself, the point

$$c = (1, 2) \tag{13}$$

is canonical at $N = 3$, and

$$\frac{0}{1} < \frac{1}{2} < \frac{2}{3}. \tag{14}$$

The new regression theorem counterexample_has_listed_mediator_without_native_decide machine-checks Equation (14) in the same local definitions. This gives a particularly sharp logical diagnosis:

$$\neg\,\mathrm{ShapeBridge} \quad \not\Rightarrow \quad \neg\,\mathrm{ListedMediatorTarget}\,. \tag{15}$$

The false step was classification through the then-current shape union. The failure did not show that the endpoint pair lacked a mediator.

Subsequent Branch-C work developed a direct constructor route, culminating locally in BC-055O. That route bypasses the refuted bridge; it does not retroactively place the counterexample inside the old shape union. Paper 10 therefore remains a valid and independently citable negative result, while the direct-constructor result belongs in a separate companion paper.

5. Why the boundary case was structurally revealing

The left endpoint $(0, 1)$ is not an arbitrary nuisance value. It exposes a mismatch between the admissible domain and the parameterization of the shape families.

The domain in Equation (6) allows zero numerators because $(0, 1)$ is a valid reduced bounded endpoint. The shape families developed before BC-054IS were centered on positive-numerator parameterizations. A universal bridge from the first domain to the second therefore required an explicit argument for the zero-left boundary. No such argument existed, and the counterexample shows that the union itself did not silently supply one.

This is a reusable formal-methods lesson:

- family theorems establish implications conditional on family membership;
- a classifier establishes that every input belongs to a listed family;
- a target theorem may sometimes be proved without any classifier;
- these three proof roles should have different theorem names and different dependency edges.

The proof-piece method succeeded because it made the missing classifier visible before the broader target was promoted.

6. Proof-piece genealogy

The counterexample was the final step in a bounded chain rather than an isolated search result.

Leaf	Exact contribution
BC-054IK	Recorded four-plus-one arithmetic-normalization requirements.
BC-054IL	Proved the four-plus-one gap identity.
BC-054IM	Proved gap positivity under $2 \le p$.
BC-054IN	Proved the order / determinant bridge from the gap identity and lower bound.
BC-054IO	Assembled the mod-3 arithmetic reducer, including the $p = 3t + 2$ gcd discharge.
BC-054IP	Proved residue exhaustiveness and classified `fourPlusOneConditions p`.
BC-054IQ	Routed each four-plus-one parameter to obstruction or an endpoint package.
BC-054IR	Named the missing arbitrary endpoint-shape bridge.
BC-054IS	Refuted that bridge with Equation (1).

BC-054IQ is a family classifier over the four-plus-one parameter p. It is not a classifier over every pair satisfying Equation (6). BC-054IR made the missing domain promotion explicit. BC-054IS then established that the promotion was false as written.

7. Lean certification and strengthened replay

Lean 4 is an interactive theorem prover and programming language based on dependent type theory [2]. The original BC-054IS source is preserved byte-for-byte in the release bundle. The copied source has SHA-256:

3C9C24EE0A3478ADBECC0DB15AD5D47383A7075E8FF020D21C240248D58D3469.

It is a standalone Lean file with no import lines. The release pins:

- Lean toolchain: leanprover/lean4:v4.30.0;
- Lean commit: d024af099ca4bf2c86f649261ebf59565dc8c622.

The historical source uses `native_decide` for several finite arithmetic checks. It compiles successfully, but an axiom report then names generated native-decision axioms for some historical theorems. Rather than hiding that dependency or rewriting the historical source, this revision adds Paper10Regression.lean, which re-proves the four reader-facing facts without `native_decide`:

Regression theorem	Checked statement
counterexample_canonical_without_native_decide	Equation (6) holds for Equation (1).
counterexample_not_shape_without_native_decide	Equation (11) holds.
endpoint_shape_bridge_refuted_without_native_decide	Equation (12) holds.
counterexample_has_listed_mediator_without_native_decide	Equations (13)-(14) hold.

All four regression theorems compile under the pinned toolchain. Their `#print axioms` reports contain only Lean's standard `propext` and `Quot.sound`; they contain no generated native-decision axiom, user-declared `axiom`, `sorry`, `admit`, or `unsafe`.

This two-layer design is source-faithful:

1. the original formal artifact remains immutable;
2. its exact hash and historical dependency surface remain visible;
3. the stronger replay layer is additive and separately hashed;
4. the new layer proves the same central result and an additional scope-regression fact;
5. no later theorem is backdated into the original artifact.

8. Repair routes

The counterexample leaves three mathematically distinct repair strategies.

8.1 Restrict the bridge domain

One can require a positive left numerator or an interior Farey interval. This would remove the counterexample from the bridge's premise. Such a restriction is legitimate only if the intended downstream theorem also has that restricted domain.

8.2 Expand the taxonomy

One can add a zero-left family. Candidate boundary chains may begin with $(0, 1)$ and pairs such as $(k, k+1)$, but this paper does not claim that a particular parameterization is exhaustive. A repaired taxonomy needs its own family conditions, construction, coverage proof, and regression tests.

8.3 Bypass the taxonomy

One can prove mediator existence directly from the canonical determinant-greater-than-one premise. This route avoids the need to classify every endpoint by a shape family. The later BC-055O work takes this form. Logically, it supplements the present result: it proves a different proposition through a different dependency chain.

9. Exact scope

The verified result is a finite theorem about one explicitly defined bridge. It establishes:

- the concrete premise in Equation (6);
- exclusion from the then-current endpoint-shape union;
- negation of the universal bridge in Equation (7);
- existence of a listed mediator for the same counterexample;
- the logical separation in Equation (15).

It does not establish a theorem about an infinite operator, a spectral identity, finite C-22, HSIT, a SIT / Riemann bridge, the Riemann Hypothesis, or full or physical Super Information Theory. Those propositions have different definitions and proof obligations. They do not follow from the local counterexample or from the direct mediator construction.

10. Reproducibility

The release bundle contains:

- the exact BC-054IS Lean source;

- the native-decision-free regression source;
- the pinned Lean toolchain;
- the original claim signature and compatibility edge;
- a SHA-256 proof manifest;
- a machine-readable replay report;
- the final manuscript and PDF;
- a release manifest covering every deposited file.

The deterministic replay performs four checks:

1. verify every governed source and evidence hash;
2. reject `sorry`, `admit`, user-declared `axiom`, or `unsafe` in the Lean sources;
3. compile the historical source under Lean 4.30.0;
4. compile the regression source and record its complete axiom report.

The formal replay result is `PASS`. The result should be interpreted at the exact theorem scope stated in Sections 3-4.

11. Conclusion

The central lesson is simple: valid family proofs do not supply a universal classifier for free. In the Branch-C endpoint program, the admissible pair $N = 3$, $(0, 1)$, $(2, 3)$ passes the determinant-greater-than-one premise and falls outside the then-current shape union. Lean therefore refutes the bridge exactly as written.

The result became more informative once the stronger target was separated from the failed route. The same pair has a listed mediator, and later formal work develops a direct constructor. The counterexample was not a dead end; it removed a false dependency and forced the proof program onto a sounder path.

References

1. A. O. Matveev, *Farey Sequences: Duality and Maps Between Subsequences*, De Gruyter, Berlin / Boston, 2017. doi:10.1515/9783110547665.
2. L. de Moura and S. Ullrich, “The Lean 4 Theorem Prover and Programming Language,” in *Automated Deduction - CADE 28*, Lecture Notes in Computer Science 12699, pp. 625-635, 2021. doi:10.1007/978-3-030-79876-5_37.

Data, code, and artifact availability

The initial public record is doi:10.5281/zenodo.21231896. The revision package adds the exact Lean source, regression source, toolchain pin, proof manifest, replay result, and SHA-256 release manifest. A public code URL should be added to the record only after the exact release directory is deposited or mirrored.

Conflicts of interest

The author declares no competing interests.